CLOSING THE LOOP: AN APPROACH TO URBAN ISSUE RESOLUTION

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Project titled "**CLOSING THE LOOP: AN APPROACH TO URBAN ISSUE RESOLUTION**" is the Bonafide work of "**PRIYANKA E (231901038) & RENUGA DEVI K (231901041)**" who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

Closing the Loop: An Approach to Urban Issue Resolution is a smart digital platform designed to improve communication between citizens and government authorities. Cities face common issues such as damaged roads, garbage overflow, water supply interruptions, and faulty street lights, which often remain unresolved due to inefficient reporting systems. This project introduces a mobile application that allows citizens to easily report issues by uploading images, descriptions, and location details. The complaints are then processed through a centralized system and displayed on a web-based dashboard for administrators. Authorities can assign, track, and update complaints until they are resolved. The system ensures transparency by providing real-time updates and enables accountability through proof-based resolution and citizen feedback. Built using Flutter, React.js, Node.js, and PostgreSQL, the platform provides a scalable and efficient solution for urban issue management. Overall, the system enhances public service delivery, reduces delays, and strengthens trust between citizens and authorities.

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CHAPTER 1

INTRODUCTION

1.1 GENERAL

Urban areas often face problems such as damaged roads, garbage overflow, water leakage, faulty street lights, and poor sanitation facilities. In many cases, citizens find it difficult to report these issues due to the lack of a proper communication platform between the public and government authorities. Traditional complaint systems rely on manual methods such as phone calls, written applications, or office visits, which are time-consuming and inefficient. Due to the absence of a centralized complaint management system, complaints are often delayed, misplaced, or unresolved, leading to poor service delivery and reduced public trust.

The Smart Complaint Management System is developed to provide a modern and efficient solution for handling urban complaints. The system consists of a mobile application for citizens and a web-based dashboard for administrators and departments. Citizens can report issues by uploading images, entering descriptions, and sharing location details directly through the application. Authorities can monitor, assign, and resolve complaints using a centralized platform, ensuring transparency and faster issue resolution.

The application is developed using Flutter for the mobile frontend, React.js for the administrative dashboard, Node.js for backend services, and PostgreSQL for database management. Real-time tracking and status updates improve communication between citizens and authorities while ensuring accountability in complaint resolution.

1.2 SCOPE OF THE WORK

This project focuses on the design and development of a Smart Complaint Management System that enables efficient reporting, tracking, and management of urban issues through digital platforms. The scope of the work includes the development of a citizen mobile application and a web-based administrative dashboard for complaint handling and monitoring.

The system allows citizens to report complaints related to roads, sanitation, water supply, electricity, and waste management by uploading supporting images, adding descriptions, and sharing GPS location details. The complaints are stored in a

centralized PostgreSQL database, enabling easy access and organized management by authorities.

The administrative dashboard provides features for viewing complaints, assigning tasks to departments, updating complaint status, and verifying resolutions. Real-time tracking functionality ensures that citizens remain informed about complaint progress from submission to resolution. The project also includes proof-based verification, where departments upload evidence after resolving issues to improve transparency and accountability.

Additionally, the system supports data analysis and visualization to identify recurring issues and affected areas, helping authorities improve decision-making and urban infrastructure planning. The project is designed with scalability, usability, and responsiveness in mind, making it suitable for smart city environments and future enhancements.

1.3 AIM AND OBJECTIVES OF THE PROJECT

The main aim of this project is to develop a Smart Complaint Management System that provides an efficient and user-friendly platform for citizens to report urban issues and for authorities to manage and resolve complaints effectively. The system is designed to improve communication between citizens and government departments by replacing traditional manual complaint methods with a digital solution that ensures transparency, accountability, and faster service delivery. The project focuses on creating a mobile application where users can submit complaints with images, descriptions, and location details, while administrators can monitor and assign complaints through a centralized web dashboard. The objectives of the project include implementing real-time complaint tracking, enabling proof-based verification after issue resolution, and maintaining all complaint data securely in a centralized database. Additionally, the system aims to analyze complaint data to identify recurring urban problems and support better decision-making for city management. Overall, the project seeks to enhance public grievance handling using modern technologies such as Flutter, React.js, Node.js, and PostgreSQL, while ensuring scalability, efficiency, and ease of use.

CHAPTER 2

LITERATURE SURVEY

Various research studies have focused on developing smart complaint management systems to improve communication between citizens and authorities. These systems aim to replace traditional complaint methods with digital platforms that provide faster reporting, centralized management, and better transparency.

Walwadkar et al. (2022) proposed a Smart Civic Issue Reporting System using Android, NLP, and machine learning techniques for complaint classification and urgency detection. The system allowed users to upload images and location details for accurate reporting. Ullah et al. (2023) developed an e-Complaint system using PHP, Laravel, and MySQL for online complaint registration and status tracking. The study highlighted the importance of centralized complaint management in improving efficiency.

Kamble et al. (2024) introduced CitizenConnect, a mobile-based platform that supports real-time complaint submission and monitoring. Reethika et al. (2025) designed a Smart City Complaint Management System using Flask, React, NLP, and IoT technologies to provide intelligent complaint prioritization and geospatial analysis. Musawarman et al. (2024) developed a Campus Complaint System with multimedia support, GPS integration, and an administrative dashboard for efficient issue management.

From the literature survey, it is observed that existing systems improve complaint handling through digital reporting and centralized monitoring. However, many systems still face limitations such as lack of offline support, limited scalability, privacy concerns, and absence of intelligent analysis features. The proposed Smart Complaint Management System addresses these challenges by providing a user-friendly mobile application, centralized dashboard, real-time tracking, proof-based verification, and efficient complaint management using Flutter, React.js, Node.js, and PostgreSQL technologies.

CHAPTER 3

SYSTEM DESIGN

3.1 ARCHITECTURE DIAGRAM

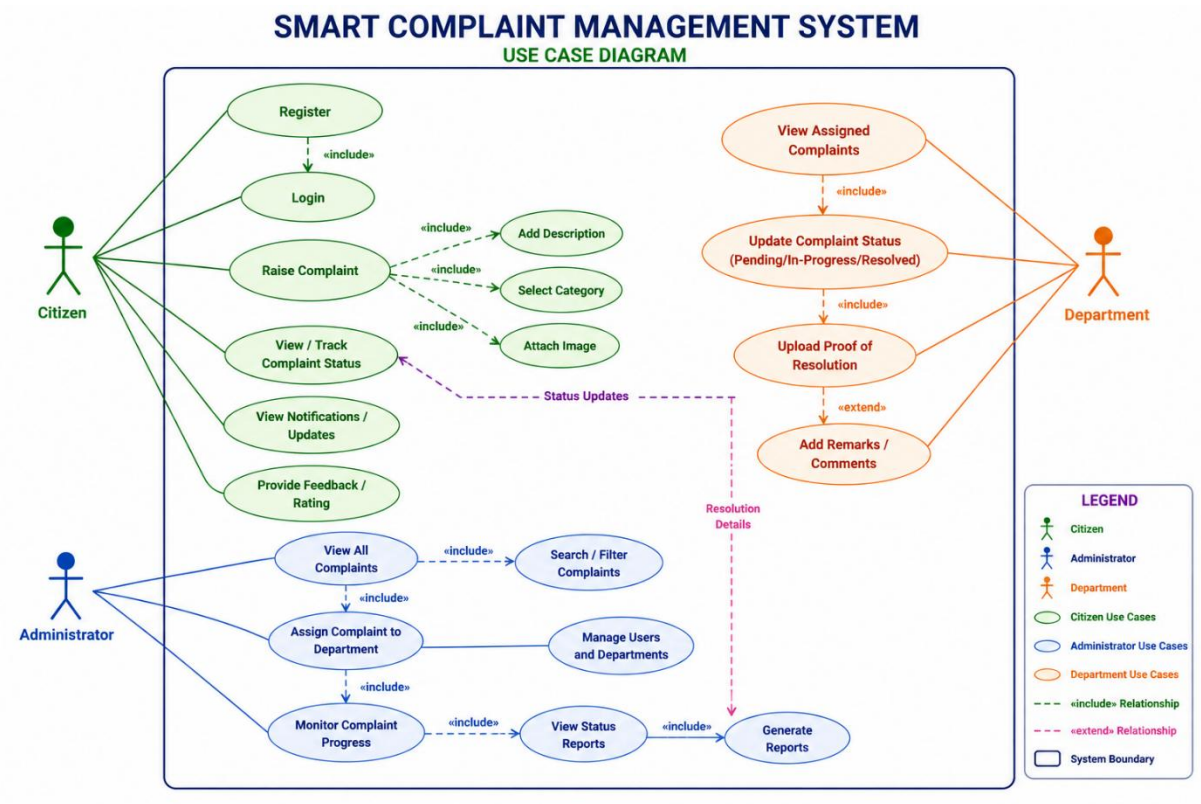
The architecture of the Smart Complaint Management System follows a client-server model consisting of a mobile application, backend server, centralized database, and administrative dashboard. Citizens use the Flutter-based mobile application to submit complaints with images, descriptions, and GPS location details. The requests are processed through the Node.js backend server, which manages complaint handling, authentication, and communication between the application and database.

All complaint data is stored securely in a PostgreSQL database for efficient management and retrieval. Administrators and departments access the system through a React.js web dashboard, where complaints can be viewed, assigned, tracked, and resolved. Real-time updates ensure transparency and smooth communication between citizens and authorities. The modular architecture improves scalability, performance, and maintainability of the system.

3.2 USE CASE DIAGRAM

The Use Case Diagram of the Smart Complaint Management System describes the interactions between citizens, administrators, and departments. Citizens can register/login, raise complaints, upload images, track complaint status, and provide feedback after resolution. Administrators can view complaints, assign tasks to departments, monitor progress, and generate reports. Departments can update complaint status, upload proof of resolution, and manage assigned complaints.

The use case design ensures proper coordination between all users and provides a structured workflow for complaint management and resolution.



3.3 HARDWARE SPECIFICATIONS

The Smart Complaint Management System is designed to run on standard mobile and computer devices without requiring specialized hardware. The recommended hardware specifications are as follows:

- Processor: Quad-core processor or higher

- RAM: Minimum 4 GB
- Storage: 64 GB internal storage or higher
- Display: Touchscreen smartphone or monitor
- Operating System: Android 8.0 or above for mobile devices, Windows/Linux for web dashboard
- Internet Requirement: Stable internet connection for real-time communication and synchronization

These specifications ensure smooth functioning of the mobile application, backend services, and administrative dashboard.

3.4 SOFTWARE SPECIFICATIONS

The Smart Complaint Management System is developed using modern web and mobile technologies to ensure efficiency, scalability, and user-friendly interaction. The software requirements are listed below:

- Mobile Application: Flutter
- Web Dashboard: React.js
- Backend Framework: Node.js and Express.js
- Database: PostgreSQL
- Programming Languages: Dart, JavaScript
- IDE/Tools: Android Studio, VS Code
- Version Control: Git and GitHub
- Operating System: Android, Windows/Linux

The combination of these technologies enables real-time complaint handling, centralized data management, responsive user interfaces, and secure communication between users and authorities.

CHAPTER 4

PROPOSED SYSTEM

The proposed Smart Complaint Management System is designed to provide an efficient and transparent platform for reporting and managing urban issues. The system replaces traditional manual complaint methods with a digital solution that improves communication between citizens and authorities while ensuring faster complaint resolution and better service management.

The system includes a mobile application developed using Flutter, where citizens can register, log in, and submit complaints by uploading images, adding descriptions, selecting categories, and sharing location details. Users can also track complaint status in real time and receive notifications regarding updates and resolutions.

A web-based administrative dashboard developed using React.js allows authorities to view complaints, assign them to the appropriate departments, monitor complaint progress, and generate reports. Departments can access assigned complaints, update status, upload proof of resolution, and add remarks after completing the work.

The backend system is implemented using Node.js and Express.js, while PostgreSQL is used for secure storage and management of complaint data. The proposed system ensures transparency, accountability, and efficient complaint handling. In addition, complaint data can be analyzed to identify frequently affected areas and recurring urban issues, helping authorities improve decision-making and city management.

CHAPTER 5

MODULE DESCRIPTION

The Smart Complaint Management System is divided into different modules to ensure efficient complaint reporting, tracking, and management. Each module is designed to perform specific functions that improve communication between citizens and authorities and ensure smooth complaint handling.

1. User Authentication Module

This module allows citizens, administrators, and departments to register and log in securely into the system. Authentication ensures that only authorized users can access the respective features and services.

2. Complaint Registration Module

The Complaint Registration Module enables citizens to submit complaints related to urban issues such as damaged roads, garbage overflow, water leakage, and street light problems. Users can upload images, add descriptions, select categories, and share location details for accurate reporting.

3. Complaint Tracking Module

This module allows users to track the status of their complaints in real time. Complaints move through stages such as Pending, In Progress, and Resolved, helping citizens stay informed about updates and resolution progress.

4. Administrative Dashboard Module

The Administrative Dashboard helps authorities manage complaints efficiently. Administrators can view all complaints, filter complaint details, assign complaints to departments, monitor complaint progress, and generate reports for analysis.

5. Department Management Module

This module is used by departments to manage assigned complaints. Departments can update complaint status, upload proof of resolution, and add remarks after resolving the issue, ensuring accountability and transparency.

6. Notification and Feedback Module

The system provides notifications and updates to citizens regarding complaint progress and resolution. After the issue is resolved, users can verify the solution and provide feedback or ratings to improve service quality.

5.1 KEY BENEFITS OF THE PROPOSED SYSTEM

Improved Complaint Management

The system provides a centralized platform for managing complaints efficiently, reducing delays and improving coordination between departments.

Real-Time Tracking

Citizens can track complaint status in real time, ensuring transparency and reducing the need for repeated follow-ups.

Better Communication

The platform improves communication between citizens and authorities through notifications, updates, and feedback mechanisms.

Increased Accountability

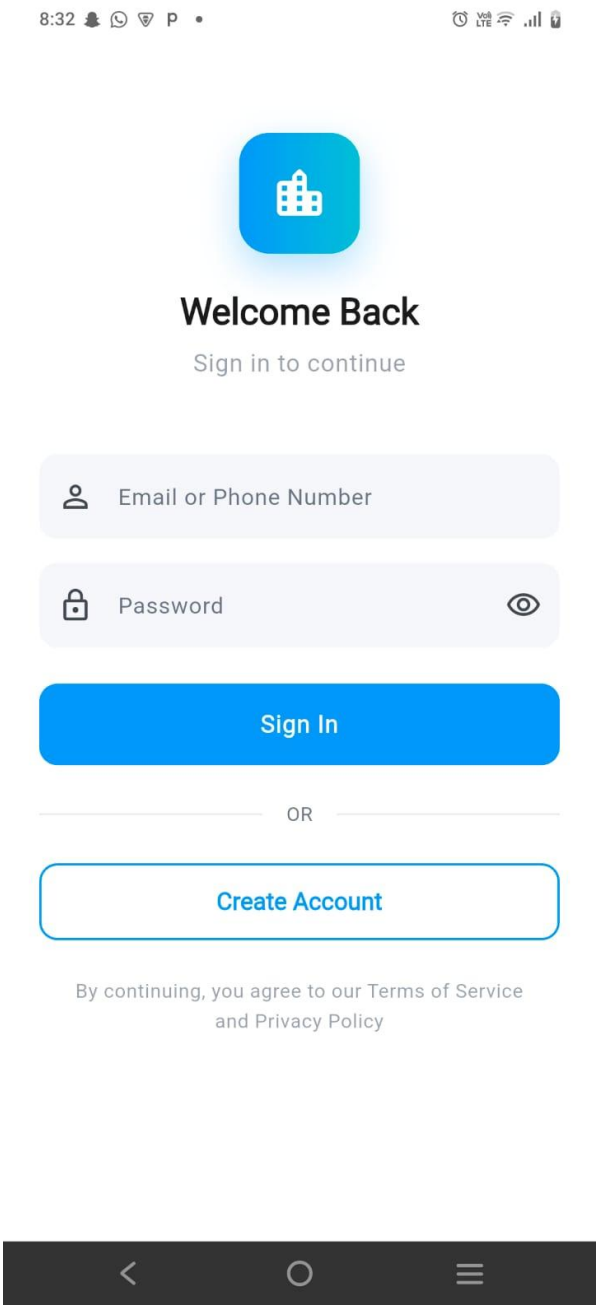
Departments upload proof of resolution after fixing issues, ensuring proper verification and accountability.

Data Analysis and Smart Management

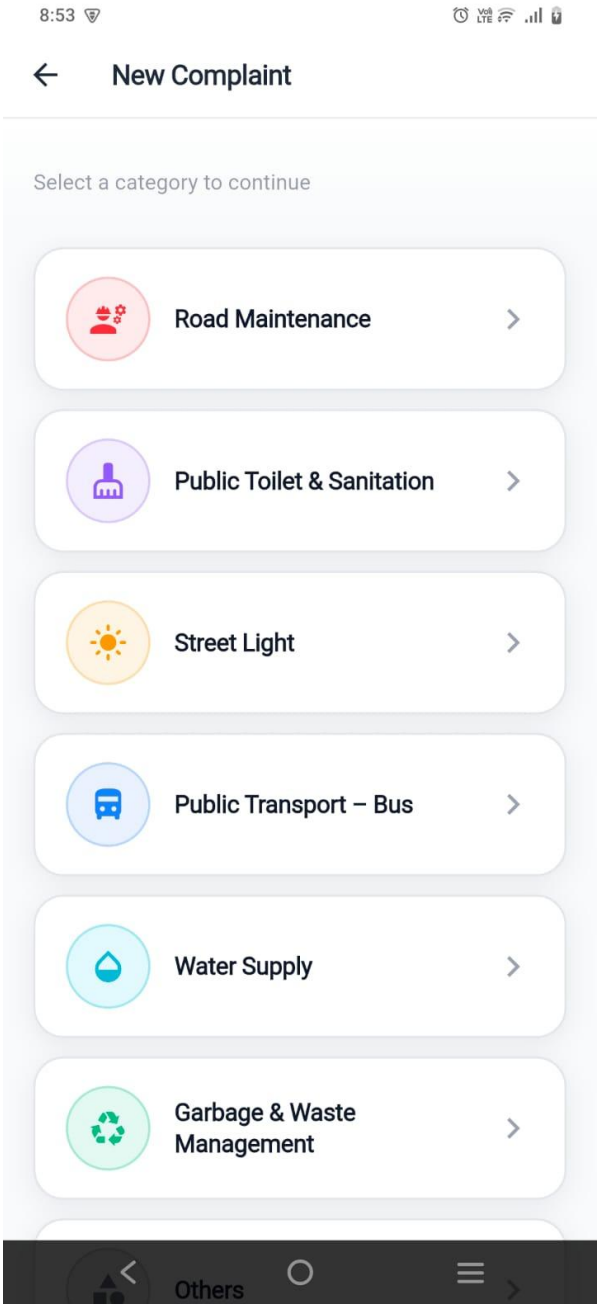
Complaint data can be analyzed to identify recurring problems and affected areas, helping authorities improve decision-making and urban management.

CHAPTER 6

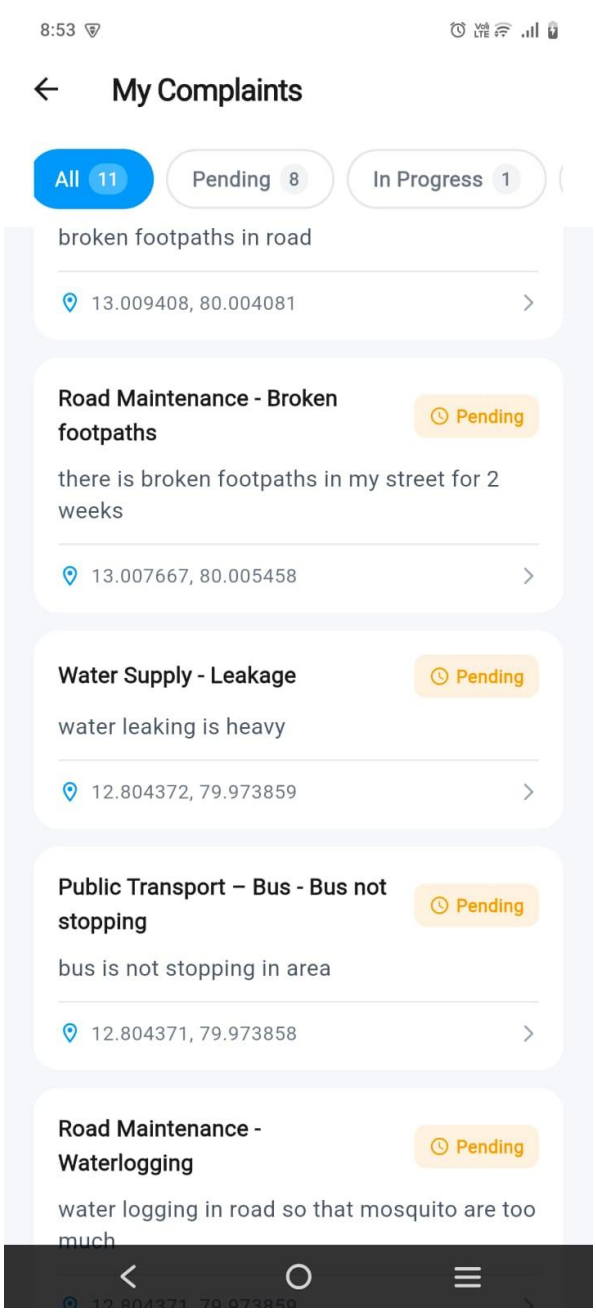
OUTPUT



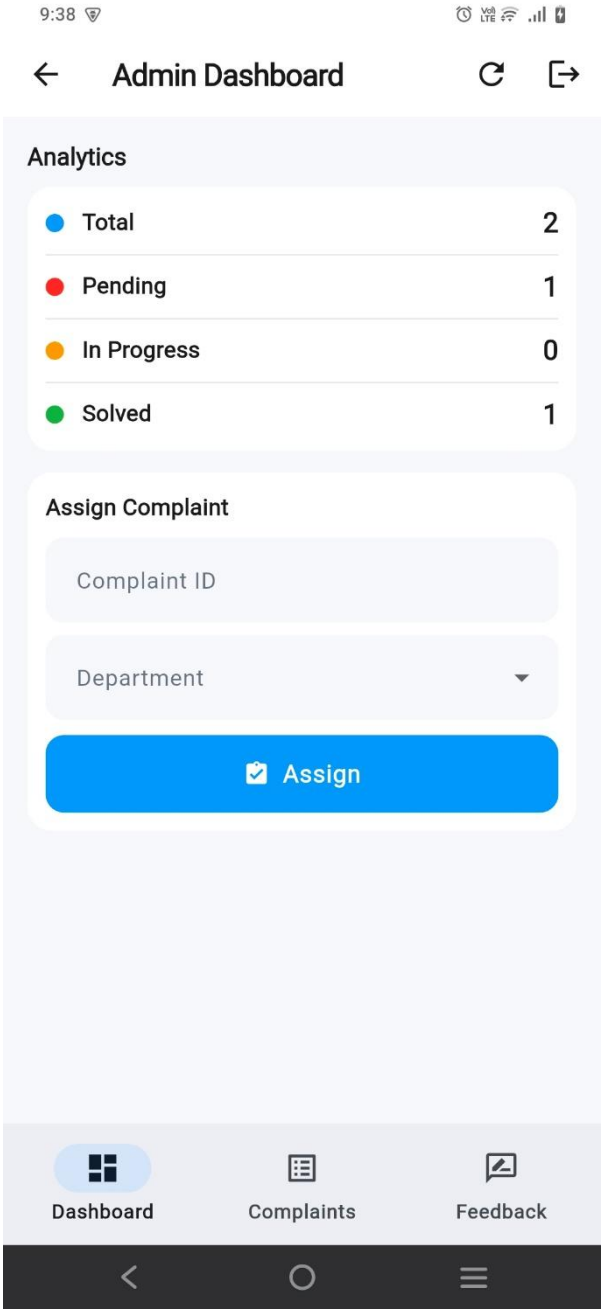
Login Page



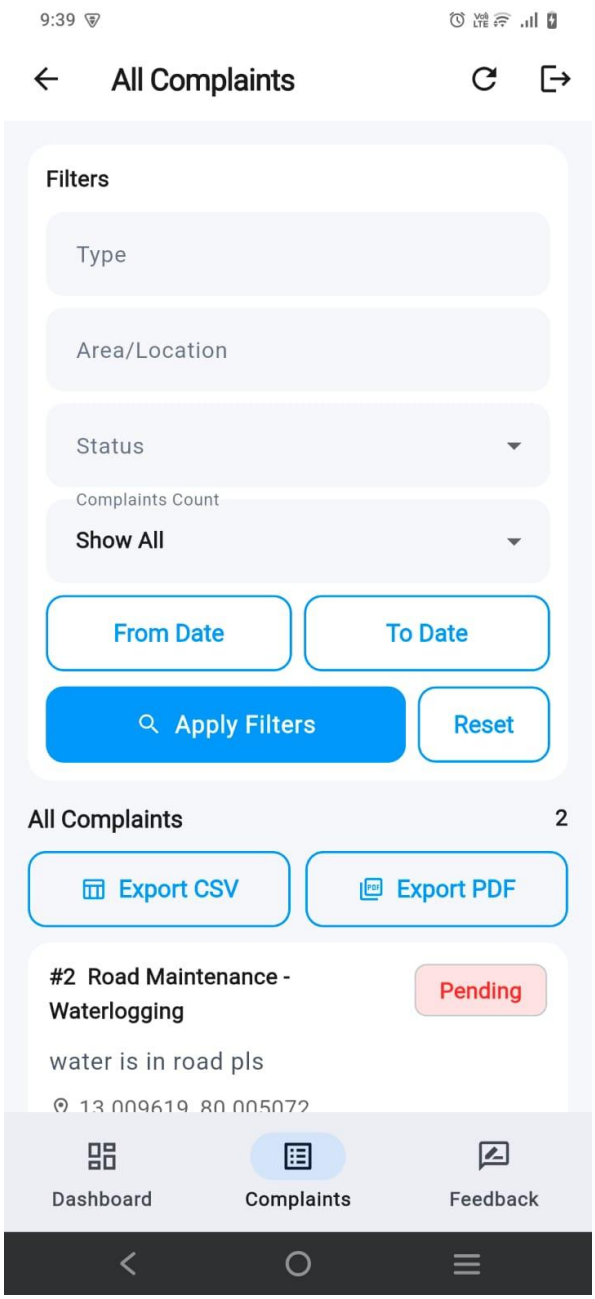
Raise Complaint Page



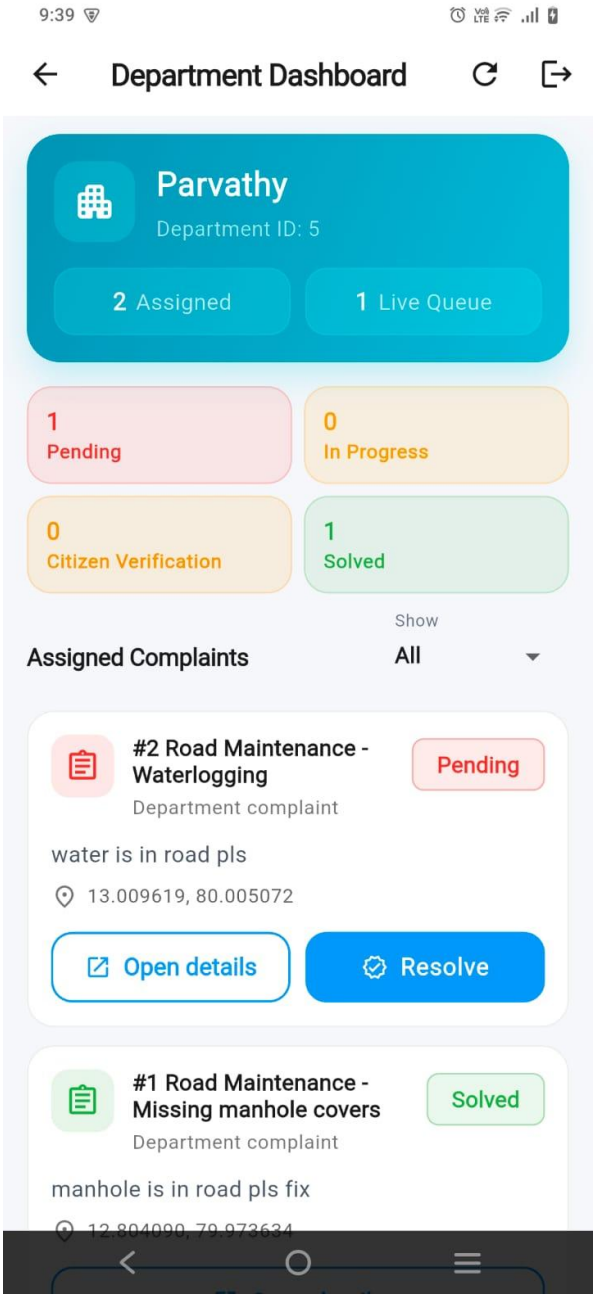
My Complaint Page



Admin Dashboard



All Complaint Page



Department Dashboard

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

The Smart Complaint Management System provides an efficient and user-friendly solution for managing urban complaints through a digital platform. The system simplifies the complaint registration process by allowing citizens to report issues using a mobile application with images, descriptions, and location details. Real-time complaint tracking, centralized management, and proof-based verification improve transparency, accountability, and communication between citizens and authorities.

The web-based administrative dashboard helps authorities monitor complaints, assign tasks to departments, and manage issue resolution effectively. By integrating modern technologies such as Flutter, React.js, Node.js, and PostgreSQL, the system ensures scalability, responsiveness, and secure data management. Overall, the project improves public grievance handling and contributes toward smarter and more efficient urban management.

Future Enhancements

- **Cybersecurity and Data Protection:** Advanced security features such as data encryption, secure authentication, and role-based access control can be implemented to improve system security and protect user data from unauthorized access.
- **Multi-Language Support:** Adding multiple language options will improve accessibility and make the application easier to use for people from different regions and language backgrounds.
- **Advanced Data Analytics:** Data analysis and visualization features can be enhanced to identify recurring urban problems, analyze complaint patterns, and support better decision-making for efficient urban management.

CHAPTER 8

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